MATH4041/MCSC6120 Assignment 1

Glucose Control, Part 1

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1. Derivation of the Two-Degree-of-Freedom Model

We begin with the glucose control model introduced in lecture, which consists of an integral equation and a differential equation:

$$u = \alpha v + \beta \int_{-\infty}^{t} \lambda e^{-\lambda(t-t')} v(t')\, dt', \tag{1}$$

and

$$\dot{v} = -\gamma v - \delta u(v + \bar{v}) + F(t). \tag{2}$$

Our goal is to eliminate the integral so that the model can be expressed as a system of two ordinary differential equations. To do this, we first differentiate the equation for u with respect to time:

$$\dot{u} = \alpha \dot{v} + \beta\lambda \frac{d}{dt}\left(\int_{-\infty}^{t} e^{-\lambda(t-t')} v(t')\, dt' \right). \tag{3}$$

Since the upper limit of integration depends on t, we apply Leibniz's rule:

$$\dot{u} = \alpha \dot{v} + \beta\lambda \left(v(t) - \lambda \int_{-\infty}^{t} e^{-\lambda(t-t')} v(t')\, dt' \right) \tag{4}$$

$$= \alpha \dot{v} + \beta\lambda v - \lambda \left(\beta \int_{-\infty}^{t} \lambda e^{-\lambda(t-t')} v(t')\, dt' \right). \tag{5}$$

Notice that the remaining integral is the same one that appears in our original equation for u. Rearranging that equation gives:

$$\beta \int_{-\infty}^{t} \lambda e^{-\lambda(t-t')} v(t')\, dt' = u - \alpha v. \tag{6}$$

We can now substitute this expression into our derivative:

$$\dot{u} = \alpha \dot{v} + \beta\lambda v - \lambda(u - \alpha v) \tag{7}$$

$$= \alpha \dot{v} + \lambda(\alpha + \beta) v - \lambda u. \tag{8}$$

Finally, we introduce the notation:

$$\dot{v} = f(u, v). \tag{9}$$

This allows us to express the original model entirely in terms of two differential equations:

$$\boxed{\begin{aligned} \dot{u} &= -\lambda u + \lambda(\alpha + \beta) v + \alpha f(u, v), \\ \dot{v} &= -\gamma v - \delta u(v + \bar{v}) + F(t). \end{aligned}} \tag{10}$$

We have therefore obtained the required two-degree-of-freedom ODE system without the integral

term.

2. Scaling the Variables

We now want to simplify the system by introducing dimensionless variables, reducing the number of parameters appearing in the equations.

Recall the system obtained in Question 1:

$$\begin{aligned} \dot{u} &= -\lambda u + \lambda(\alpha+\beta)v + \alpha f(u,v), \\ \dot{v} &= -\gamma v - \delta u(v+\bar{v}) + F(t). \end{aligned} \tag{11}$$

Following the scaling approach from lecture, we introduce:

$$t = \tilde{t}\,t', \qquad u = \tilde{u}\,u', \qquad v = \bar{v}\,v'. \tag{12}$$

Because time is also being rescaled, we must account for this when transforming the derivatives:

$$\dot{u} = \frac{\tilde{u}}{\tilde{t}}\frac{du'}{dt'}, \qquad \dot{v} = \frac{\bar{v}}{\tilde{t}}\frac{dv'}{dt'}. \tag{13}$$

Here, $f'(u', v')$ represents the original dimensional function f evaluated using the scaled variables.

Substituting the new variables into both equations gives:

$$\frac{\tilde{u}}{\tilde{t}}\frac{du'}{dt'} = -\lambda\tilde{u}u' + \lambda(\alpha+\beta)\bar{v}v' + \alpha f'(u',v'), \tag{14}$$

$$\frac{\bar{v}}{\tilde{t}}\frac{dv'}{dt'} = -\gamma\bar{v}v' - \delta\tilde{u}u'(\bar{v}v'+\bar{v}) + F(t). \tag{15}$$

Next, we isolate the derivatives by dividing through by their respective coefficients:

$$\frac{du'}{dt'} = -(\lambda\tilde{t})u' + \left(\frac{\lambda\bar{v}\tilde{t}}{\tilde{u}}\right)(\alpha+\beta)v' + \left(\frac{\alpha\tilde{t}}{\tilde{u}}\right)f'(u',v'), \tag{16}$$

$$\frac{dv'}{dt'} = -(\gamma\tilde{t})v' - (\delta\tilde{u}\tilde{t})u'(v'+1) + \frac{\tilde{t}}{\bar{v}}F(t). \tag{17}$$

To make the second equation simpler, we choose the time and u-scales such that:

$$\gamma\tilde{t} = 1, \qquad \delta\tilde{u}\tilde{t} = 1. \tag{18}$$

This gives:

$$\boxed{\tilde{t} = \frac{1}{\gamma}, \qquad \tilde{u} = \frac{1}{\delta\tilde{t}} = \frac{\gamma}{\delta}.} \tag{19}$$

We also introduce the scaled input parameter:

$$\boxed{G = \frac{\tilde{t}}{\bar{v}}F(t).} \tag{20}$$

With these choices, the system becomes:

$$\frac{du'}{dt'} = -\frac{\lambda}{\gamma}u' + \frac{\lambda\delta\bar{v}}{\gamma^2}(\alpha+\beta)v' + \frac{\alpha\delta}{\gamma^2}f'(u',v'), \tag{21}$$

$$\frac{dv'}{dt'} = -v' - u'(v'+1) + G. \tag{22}$$

We can simplify the notation further by defining three new parameters:

$$\boxed{a = \frac{\lambda}{\gamma}, \qquad b = \frac{\lambda\delta\bar{v}}{\gamma^2}(\alpha+\beta), \qquad c = \frac{\alpha\delta}{\gamma^2}.} \tag{23}$$

Using these definitions, the dimensionless system can be written more compactly as:

$$\boxed{\begin{aligned}\frac{du'}{dt'} &= -au' + bv' + cf'(u',v'),\\ \frac{dv'}{dt'} &= -v' - u'(v'+1) + G.\end{aligned}} \tag{24}$$

This completes the scaling process, leaving us with a simpler system involving the reduced parameters a, b, c, and the input G.

3. Equilibrium Solutions

To determine the equilibrium solutions, we set both time derivatives equal to zero. Since our equations are now dimensionless, we can drop the primes for simplicity.

The equilibrium conditions are:

$$\boxed{\begin{aligned}-au + bv &= 0,\\ -v - u(v+1) + G &= 0.\end{aligned}} \tag{25}$$

At equilibrium, $f(u,v) = 0$ because it represents the v-equation, whose derivative must also vanish.

We begin by solving the first equation for u:

$$-au + bv = 0. \tag{26}$$

Rearranging gives:

$$\boxed{u^* = \frac{b}{a}v^*.} \tag{27}$$

We then substitute this relationship into the second equilibrium equation:

$$-v - \frac{b}{a}v(v+1) + G = 0. \tag{28}$$

Expanding and collecting terms:

$$-\frac{b}{a}v^2 - \frac{b}{a}v - v + G = 0, \tag{29}$$

$$v^2 + \left(1 + \frac{a}{b}\right) v - \frac{a}{b}G = 0. \tag{30}$$

This is a quadratic equation in v, so applying the quadratic formula gives:

$$\boxed{v^* = -\frac{1}{2}\left(1 + \frac{a}{b}\right) \pm \frac{1}{2}\sqrt{\left(1 + \frac{a}{b}\right)^2 + \frac{4a}{b}G.}} \tag{31}$$

The number of real equilibrium solutions depends on the expression inside the square root. In particular, the two solutions coincide when:

$$\left(1 + \frac{a}{b}\right)^2 + \frac{4a}{b}G = 0. \tag{32}$$

Solving this equation for G gives the critical saddle-node value:

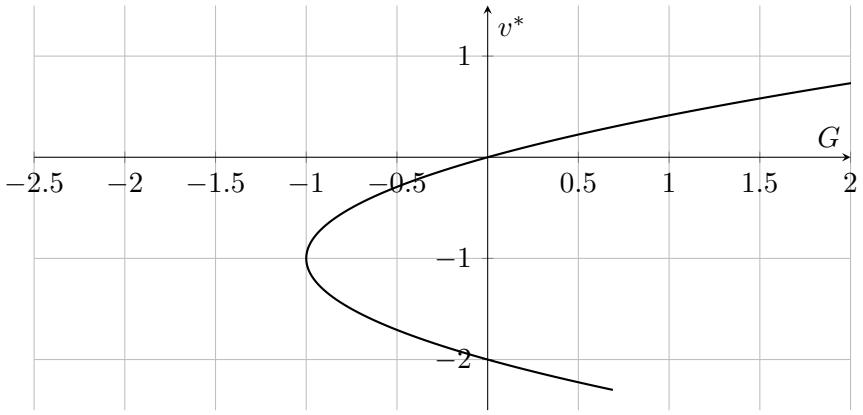
$$\boxed{G_{\text{SN}} = -\frac{b}{4a}\left(1 + \frac{a}{b}\right)^2.} \tag{33}$$

Assuming $a > 0$ and $b > 0$, this critical value separates the three possible cases. For $G < G_{\text{SN}}$, there are no real equilibria. At $G = G_{\text{SN}}$, the two solutions merge into one, while for $G > G_{\text{SN}}$, two distinct real equilibria exist.

We can summarize these results in the following table:

Equilibrium Solutions as a Result of G		
Value of G	**Number of Solutions**	**Equilibrium Solution(s)**
$G < G_{\text{SN}}$	Zero	N/a
$G = G_{\text{SN}}$	One	$(u^*, v^*) = \left(\frac{b}{a}v^*, -\frac{1}{2}\left(1 + \frac{a}{b}\right)\right)$
$G > G_{\text{SN}}$	Two	$(u^*, v^*) = \left(\frac{b}{a}v^*, -\frac{1}{2}\left(1 + \frac{a}{b}\right) \pm \frac{1}{2}\sqrt{\left(1 + \frac{a}{b}\right)^2 + \frac{4a}{b}G}\right)$

To visualize these results, we can sketch the equilibrium branch against the input parameter G. The curve illustrates how the two equilibrium solutions meet at the saddle-node point.



As shown in the sketch, the equilibrium branch bends at the critical value G_{SN}. This is where the two real equilibrium solutions coincide, illustrating the saddle-node behaviour of the model.